BREAST CARE-MA DIAGNOSTIC DECISION SUPPORT REPORT

Department of Information Technology | Nehru Institute of Engineering and Technology
Explainable Multi-Agent AI System for Ultrasound Modality

Case ID:	TEST-BENIGN-01	**Date & Time:**	2026-10-31 00:06:33
Patient Age:	48	**Family History:**	No
Breast Density:	B - Scattered fibroglandular	**Symptoms:**	Routine mammography screening

1. Quantitative AI Findings & Model Outputs

Parameter	Result	Clinical Meaning
AI Classification	**BENIGN**	Calibrated Malignant Probability: 0.0%
Tumor Detection	Lesion Detected	Relative Area: 37.10%
Morphology (Agent 3)	Irregular / Spiculated	Aspect Ratio: 1.32, Solidity: 0.63
Acoustic Shadowing	Absent	Acoustic attenuation distal to mass
ACR BI-RADS Category	**BI-RADS 4B**	**Moderate Suspicion (15% - 50%)**

2. Retrieved Evidence & Clinical Guidelines (Agent 4)

• **Definitively Benign Lesion** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): Findings include simple anechoic cyst with posterior acoustic enhancement or typical benign intramammary lymph node. Zero risk of malignancy (<0%).

• **Probably Benign Lesion (<2% Malignancy Risk)** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): Solid mass with circumscribed margins, oval shape, and parallel orientation (wider than tall), most commonly fibroadenoma. Malignancy probability is <= 2%.

• **Classic Malignant Presentation (>=95% Probability)** (ACR BI-RADS Atlas 5th Edition / Society of Breast Imaging): Classic presentation: Spiculated margins, taller-than-wide (non-parallel) orientation, hypoechoic with acoustic shadowing, microcalcifications. Malignancy risk exceeds 95%.

3. Recommended Clinical Follow-up Actions

• Ultrasound-guided core needle biopsy (CNB) recommended.

• Correlate histologic findings with radiographic features to confirm concordance.

REGULATORY & SAFETY DISCLAIMER: NOTICE: Breast CARE-MA is an AI decision-support research prototype developed for academic evaluation. It does not provide definitive medical diagnosis and must never supersede clinical judgment by a licensed radiologist or oncologist.